Agent trace Illustrative replay with sample values · nothing is computed Replay

Request: "How much water is stored in the upper 5 m at this montane catchment site, and how did it change between the spring and late-summer surveys?"

Understand the request — Mid-tier · 0.9 s

Site: **a montane catchment site (sample)** · product **subsurface water storage** · electrical resistivity and electromagnetics.

Ground the request — 7 passages · 3 sources

Library docs and docstrings 2,604 | Curated catalog metadata 512 | Published studies 296 Search over 3,412 chunks; the top 7 are kept.

> *"Convert resistivity to water content using Waxman-Smits model."* `PyHydroGeophysX/petrophysics/resistivity_models.py`

Load skills — ERT | EM | Time-lapse | Petrophysics | Uncertainty — Procedures the agent reads when a task calls for them — 5 of 5 needed

Plan with structured output — Mid-tier · 1.4 s

Constrained decoding against plan schema v1.2; the plan validated on the first pass. 6 tool calls, 2 inferred values, each cited.

Call registered tools — 6 calls

`load_ert_surveys` → `invert_time_lapse` → `load_tdem_data` → `invert_tdem` → `convert_water_content` → `write_report`

Execute deterministically — Pinned environment

The validated plan runs in a pinned environment with no model in the loop, and writes the report, figures and data files with a provenance record.

Deliver **Sample values** Open the result in the workspace →

Storage, 0 to 5 m: **1,056 ± 38** mm of water, line mean Seasonal change: **−110 ± 13** mm, between the spring and late-summer surveys

✓ Curator review required After curator approval, deposit to HydroShare under CC BY 4.0, inherited from the source surveys.

Design preview. Values are illustrative and no computation is performed.